Indian Institute of Technology Kanpur

Department of Mathematics and Statistics

Analysis I (MTH 301)

End semester examination

Date: 23/11/2024

Time: 3 hours | Maximum marks: 40

Solutions

1. (a) State *Riemann's rearrangement theorem* for series of real numbers.

Solution. *Let $\sum_{n=1}^{\infty} a_n$ be a series of real numbers which is conditionally convergent and $-\infty \leqslant \alpha \leqslant \beta \leqslant +\infty$. Then there exists a rearrangement, say $\sum_{n=1}^{\infty} a'_n$, of $\sum_{n=1}^{\infty} a_n$ such that*

$$\liminf_{n\to\infty} s'_n = \alpha \text{ and } \limsup_{n\to\infty} s'_n = \beta,$$

where $\{s'_n\}_{n=1}^{\infty}$ is the sequence of partial sums of the series $\sum_{n=1}^{\infty} a'_n$.

Comments. *When a theorem has been asked to state, it has to be stated in the full generality, NOT to be specialized.*

Common mistakes.

1. *Some students have written* $\limsup_{n\to\infty} a_n$ *and* $\liminf_{n\to\infty} a_n$.
2. $\limsup$ *and* $\liminf$ *are to be considered.*
3. α, β *can be* $\pm\infty$. *Mark/s has/have been deducted for missing that.*
4. *Partial credit has been given only in case the student has mentioned all of the following cases:*
   - *(i)* $\forall \ell \in \mathbb{R}$, *there exists a rearrangement, say* $\sum_{n=1}^{\infty} a'_n$, *of* $\sum_{n=1}^{\infty} a_n$ *such that* $\sum_{n=1}^{\infty} a'_n = \ell$,
   - *(ii) there exists a rearrangement which diverges to* $+\infty$, *and*
   - *(iii) there exists a rearrangement which diverges to* $-\infty$.

(b) Does there exist a differentiable function $f : \mathbb{R} \longrightarrow \mathbb{R}$ such that f' is continuous precisely at all points of $\mathbb{R}\backslash(0,1)$? Justify your answer.

Solution. *For each $n \in \mathbb{N}$, we let $f_n : \mathbb{R} \longrightarrow \mathbb{R}$ be defined by $f_n(x) \overset{\text{def}}{=} \frac{f\left(x+\frac{1}{n}\right)-f(x)}{\frac{1}{n}}$. Clearly $f_n \xrightarrow[n\to\infty]{} f'$ pointwise on $\mathbb{R}$. Since $\mathbb{R}$ is complete, the set of all points at which f' is continuous is dense in $\mathbb{R}$. Hence it must contain a point from $(0,1)$. Therefore, no such function exists.*

Common mistakes.

1. *Some student just says, "It is a direct consequence of Baire category theorem" without any further explanations. This is NOT acceptable.*
2. *This is NOT true that the discontinuity set is always of first category. For example, the characteristic function of $\mathbb{Q}$ is discontinuous everywhere. The conclusion is true if f is the pointwie limit of a sequence of continuous functions on a complete metric space.*

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[2+2=4]

2. Let X be a metric space, and A and B are closed subsets of X. Assume that $A \cap B = \varnothing$. Show that there exists a continuous function $f : X \longrightarrow \mathbb{R}$ satisfying the following three conditions simultaneously:

   (i) $\forall x \in X, 0 \leqslant f(x) \leqslant 1$, (ii) $\forall x \in A, f(x) = 0$, and (iii) $\forall x \in B, f(x) = 1$.

[3]

Solution. *Consider the function $f : X \longrightarrow \mathbb{R}$, defined by*

$$f(x) \stackrel{\text{def}}{=} \frac{d(x,A)}{d(x,A) + d(x,B)},$$

Observe that, for all $x \in X$, $d(x,A) + d(x,B) \neq 0$, as otherwise A and B won't be disjoint. Now as $x \mapsto d(x,A)$ and $x \mapsto d(x,B)$ are continuous, so is f. (i) is obvious. Since A and B are closed, $d(x,A) = 0$ if and only if $x \in A$, and similarly for B. From this, (ii) and (iii) are clear.

Comments.

1. *It requires to show that the denominator is never zero, otherwise the above definition would not make sense. This is where one needs the assumptions that A and B are closed and disjoint.*
2. *This was Exercise 3.18 or Exercise Sheet 6, and it was also revisited once again in the class while constructing an unbounded continuous function on a noncompact metric space.*

Common mistakes.

1. *Many students missed the crucial point that the denominator cannot be* 0.
2. *Functions like* $\cos x$ *or* $\sin x$ *do not make sense on an arbitrary metric space.*
3. *X, A and B are arbitrary. They need not be subsets of $\mathbb{R}$.*

3. Find the sum of the series: $\displaystyle\sum_{n=0}^{\infty} \frac{3^n(n+1)}{n!}$. [3]

Solution (1st solution)**.** *We observe that $\sum_{n=0}^{\infty} \frac{3^n n}{n!}$ is convergent as $\sum_{n=0}^{\infty} \frac{3^n}{n!}$ is convergent. Hence we have*

$$\sum_{n=0}^{\infty} \frac{3^n(n+1)}{n!} = \sum_{n=0}^{\infty} \frac{3^n n}{n!} + \sum_{n=0}^{\infty} \frac{3^n}{n!} = 3e^3 + e^3 = 4e^3.$$

Solution (2nd solution)**.** *For all xin$\mathbb{R}$, we have $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \Longrightarrow xe^x = \sum_{n=0}^{\infty} \frac{x^{n+1}}{n!}$. Since term by term differentiation is permissible for power series, one obtains that $xe^x + e^x = \sum_{n=1}^{\infty} \frac{(n+1)x^n}{n!}$, for all $x \in \mathbb{R}$. Hence the desired sum is $4e^3$.*

Solution (3rd solution)**.** *By ratio test, it is easy to see that* $\sum_{n=0}^{\infty} \frac{3^n(n+1)}{n!}$ *converges, consequently* $\sum_{n=0}^{\infty} \frac{3^n(n+1)}{n!}x^n$ *converges for all* $x \in (-1,1)$*. Consider the function* $f : (-1,1) \longrightarrow \mathbb{R}$*, defined by*

$$f(x) \stackrel{\text{def}}{=} \sum_{n=0}^{\infty} \frac{3^n(n+1)}{n!}x^n.$$

Then f *is continuous. Since term by term integration is permitted for power series, we obtain that*

$$\int_0^x f(t)\,dt = \sum_{n=0}^{\infty} \frac{3^n}{n!}x^{n+1} = xe^{3x}, \forall x \in (-1,1).$$

Now using the fundamental theorem of calculus, we obtain that

$$f(x) = 3xe^{3x} + e^{3x}, \forall x \in (-1,1).$$

Now by Abel's limit theorem, we get

$$\sum_{n=1}^{\infty} \frac{3^n(n+1)}{n!} = \lim_{x\to 1-} f(x) = 4e^3.$$

Comments. *This is the most easiest question in this paper which is expected to be answered by everyone! The goal of this question was* ***to find the sum****. So you have to calculate the sum. Partial credits will be awarded only if for some silly error the final answer has not been calculated right, or some justification has not been provided in some intermediate steps.*

Common mistakes.

1. *Even if* $\sum_{n=1}^{\infty}(a_n + b_n)$ *is convergent, it is NOT permissible to write the following right away:*

$$\sum_{n=1}^{\infty}(a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n, \tag{3.1}$$

   as $\sum_{n=1}^{\infty} a_n$ *and* $\sum_{n=1}^{\infty} b_n$ *need not converge in general. For instance, one can consider that* $a_n = 1$ *and* $b_n = -1$*, for all* $n \in \mathbb{N}$*. So, mathematically speaking, it is NOT completely correct to write* (3.1) *directly without showing (or even mentioning) beforehand that both* $\sum_{n=1}^{\infty} a_n$ *and* $\sum_{n=1}^{\infty} b_n$ *converge. Marks will be deducted for jumping the gun.*
2. *The series* $1 + 3 + 3^2 + \ldots$ *does not converge.*

4. Let $f : \mathbb{R} \longrightarrow \mathbb{R}$ be defined as $f(x) \stackrel{\text{def}}{=} \sum_{n=1}^{\infty} \frac{nx - [nx]}{n^2}$. Show that f is Riemann integrable on every closed and bounded interval. [3]

Solution. *For all* $n \in \mathbb{N}$ *and* $x \in \mathbb{R}$*, we observe that* $0 \leqslant nx - [nx] < 1$*. From Weierstrass M-test, it follows that* $\sum_{n=1}^{\infty} \frac{nx-[nx]}{n^2}$ *converges uniformly on* $\mathbb{R}$*, as* $\sum_{n=1}^{\infty} \frac{1}{n^2}$*. For all* $n \in \mathbb{N}$*, denote by* $f_n : \mathbb{R} \longrightarrow \mathbb{R}$*,* $f_n(x) \stackrel{\text{def}}{=} \frac{nx-[nx]}{n^2}$*. It is easy to see that, for all* $n \in \mathbb{N}$*,* f_n *is continuous on* $\mathbb{R}\setminus\left\{\frac{p}{n} : p \in \mathbb{Z}\right\}$*. Let* $[a,b] \subseteqq \mathbb{R}$*. Since* $[a,b] \cap \left\{\frac{p}{n} : p \in \mathbb{Z}\right\}$ *is finite, all* f_n*'s are Riemann integrable on* $[a,b] \subseteqq \mathbb{R}$*. Now from the uniform convergence, we conclude that* f *is Riemann integrable* $[a,b]$*.*

Comments.

1. *Alternatively, one can use Lebesgue criterion by showing that f is continuous at all irrational points, as so is each f_n.*
2. *Uniform convergence of the series $\sum_{n=1}^{\infty} \frac{nx-[nx]}{n^2}$ is the **most crucial** part of the argument. Without this one can neither guaranty that $D(f) \subseteqq \mathbb{Q}$ (and hence has measure zero) nor the Riemann integrability of f.*

Common mistakes.

1. *Just showing f is bounded, although that too requires that $\sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$, does not seem to help further.*
2. *Some students did not apply Weierstrass M-test correctly, for instance, considered the convergence $\frac{1}{n^2} \xrightarrow[n\to\infty]{} 0$ in place of $\sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$.*
3. *Some of you did not provide any justification why an f_n is Riemann integrable on a closed and bounded interval. It has to be clearly mentioned that on each $[a,b]$, f_n is continuous except at finitely many points.*
4. *Those who used the Riemann integrability of the greatest integer function, also need to provide a similar reason.*

5. Find all power series with real coefficients and center that converge uniformly on $\mathbb{R}$. [3]

Solution. *Let $\sum_{n=1}^{\infty} a_n(x-a)^n$ be such a power series. Then from the uniform convergence, there exists $N \in \mathbb{N}$ such that for all $m > n \geqslant N$*

$$\left| \sum_{k=n+1}^{m} a_k(x-a)^k \right| < 1, \ \forall x \in \mathbb{R}.$$

In particular, for all $n \geqslant N+1$,

$$|a_n(x-a)^n| < 1, \ \forall x \in \mathbb{R}. \tag{5.2}$$

It follows that, for all $n \geqslant N+1$, $a_n = 0$, otherwise $|a_n(x-a)^n| \xrightarrow[x\to\infty]{} \infty$ which challenges (5.2). Thus the power series reduces to a polynomial having degree at most N. Obviously polynomials are those power series that converge uniformly on $\mathbb{R}$. Hence such power series are precisely polynomials.

Common mistakes.

1. *$\sum_{n=0}^{\infty} \frac{x^n}{n!}$ does not converge uniformly on $\mathbb{R}$. The convergence is uniform only on bounded subsets of $\mathbb{R}$.*
2. *A power series need not converge uniformly on its interval of convergence. So, even if $R = \infty$, the power series need not converge uniformly on $\mathbb{R}$.*
3. *Some students wrote that such power series are the ones for which $\lim_{n\to\infty} |a_n|^{\frac{1}{n}} = 0$ (or some such things). That is nothing but same as saying $R = \infty$.*

6. Let $a < b$ and $\alpha : [a,b] \longrightarrow \mathbb{R}$ be a continuous and increasing function. Suppose that $\{f_n\}_{n=1}^{\infty}$ is a sequence of uniformly bounded functions on $[a,b]$. Assume that, for all $n \in \mathbb{N}$ $f_n \in \mathscr{R}(\alpha)$ on $[a,b]$. For all $n \in \mathbb{N}$, define $F_n : [a,b] \longrightarrow \mathbb{R}$ by

$$F_n(x) \stackrel{\text{def}}{=} \int_a^x f_n(t)\, d\alpha(t).$$

Show that $\{F_n\}_{n=1}^{\infty}$ has a subsequence which converges uniformly on $[a,b]$. [4]

Solution. *Let $M > 0$ be such that $\forall n \in \mathbb{N}$ and $x \in [a,b]$, $|f_n(x)| \leqslant M$. It is easy to see that,*

$$|F_n(x)| \leqslant \left| \int_a^x f_n\, d\alpha \right| \leqslant M(\alpha(b) - \alpha(a)),\ \forall x \in [a,b] \text{ and } n \in \mathbb{N}.$$

This shows that $\{F_n\}_{n=1}^{\infty}$ is uniformly bounded.

Let $\varepsilon > 0$. Since α is continuous and $[a,b]$ is compact, $\exists \delta > 0$ such that $|\alpha(x) - \alpha(y)| < \frac{\varepsilon}{M}$, whenever $x, y \in [a,b]$ and $|x-y| < \delta$. It is now easy to see that, if $x, y \in [a,b]$ and $|x-y| < \delta$, one has

$$|F_n(x) - F_n(y)| = \left| \int_x^y f_n\, d\alpha \right| \leqslant M|\alpha(x) - \alpha(y)| < \varepsilon, \forall n \in \mathbb{N}.$$

Hence $\{F_n\}_{n=1}^{\infty}$ is equicontinuous. Thus it admits a uniformly convergent subsequence.

Common mistakes.

1. *It cannot be concluded from $|F_n(x)| \leqslant M(\alpha(x) - \alpha(a))$ only that $\{F_n\}_{n=1}^{\infty}$ is uniformly bounded, as the RHS has to be independent of x.*
2. *Some students seemed to have picked an arbitrary sequence $\{F_{n_k}\}_{k=1}^{\infty}$ and then argued that it is convergent. There absolutely no reason for that. In fact, the mistake that they have made is $|\alpha(x) - \alpha(a)| < \varepsilon$ need not hold for all $x \in [a,b]$. Hence there is no reason to have $|F_{n_k}(x) - F_{n_m}(x)| < \varepsilon$ for all $x \in [a,b]$. Some others have argued that the original $\{F_n\}_{n=1}^{\infty}$ is uniformly convergent. This also has similar issues.*
3. *Neither f_n's are assumed to be continuous, nor α is differentiable. Hence F_n's need not be differentiable.*
4. *It has not been given that $\{f_n\}_{n=1}^{\infty}$ is uniformly or even pointwise convergent. So you cannot assume that $\{f_n\}_{n=1}^{\infty}$ converges to some f.*
5. *$C[a,b]$ is not compact. Bolzano-Weierstrass property does not apply directly.*

7. Find all continuously differentiable solutions $y : \mathbb{R} \longrightarrow \mathbb{R}$ of the following differential equation:

$$\left(\frac{dy}{dx}\right)^2 - y^2 = 0,$$

subject to the condition that $\forall x \in \mathbb{R}$, $y(x) \neq 0$. [4]

Solution (1st solution)**.** *Since y does not vanish anywhere, we obtain that $\frac{y'}{y}$ is continuous function on $\mathbb{R}$. Since $(y')^2 = y^2$, we have, for any $x \in \mathbb{R}$, $\frac{y'(x)}{y(x)} = 1$ or -1. Now from the connectedness of $\mathbb{R}$, we conclude that $\frac{y'}{y}$ must be constant. From this we obtain that all solutions are of the form Ae^x, $\forall x \in \mathbb{R}$ or Be^{-x}, $\forall x \in \mathbb{R}$, where A, B are arbitrary constants.*

Solution (2nd solution). *First we observe that, for any $t \in \mathbb{R}$, exactly one of the factors $(y'(t) - y(t))$ or $(y'(t) + y(t))$ vanishes. Otherwise, if $y'(t) - y(t) = y'(t) + y(t) = 0$, for some $t \in \mathbb{R}$, then y would vanish at that point, which is not possible. Consider the set*

$$S \stackrel{\text{def}}{=} \{t \in \mathbb{R} : y'(t) - y(t) = 0\}.$$

Since y and y' are continuous, it is clear that S is closed. We now claim that S is open. Let $t_0 \in S$. Then $y'(t_0) + y(t_0) \neq 0$. From continuity, we obtain a $\delta > 0$ such that, for all $t \in (t_0 - \delta, t_0 + \delta)$, $y'(t) + y(t) \neq 0$. This implies that $(t_0 - \delta, t_0 + \delta) \subseteqq S$. Hence S is open. Now the connectedness of $\mathbb{R}$ yields that $S = \varnothing$ or $\mathbb{R}$, i.e., one of the factors $(y' - y)$ and $(y' + y)$ must vanish identically on $\mathbb{R}$.

Comments.

1. *If the product of two functions, say f and g, vanishes identically, then it **cannot** be said in general that one of them vanishes identically. It may happen that at some f vanishes, while g might vanish at some other point. For instance, consider the functions $f, g : \mathbb{R} \longrightarrow \mathbb{R}$ defined as follows:*

$$f(x) \stackrel{\text{def}}{=} \begin{cases} x, & \text{if } x \leqslant 0; \\ 0 & \text{if } x \geqslant 0 \end{cases} \text{ and } g(x) \stackrel{\text{def}}{=} \begin{cases} 0, & \text{if } x \leqslant 0; \\ x, & \text{if } x \geqslant 0. \end{cases}$$

   Clearly $fg \equiv 0$ but none of them is identically 0.

2. *The heart of the argument is the step that exactly one of $(y' - y)$ and $(y' + y)$ vanishes identically on $\mathbb{R}$, which is due to the connectedness of $\mathbb{R}$. Unless that is established, it is NOT justified why solving the pair of $y' - y = 0$ and $y' + y = 0$ would lead us to all the solutions to the given differential equation. **This is exactly where one needs analysis**.*

3. *As far as solving $y' - y = 0$ and $y' + y = 0$ are concerned, that is a utterly trivial exercise even for a 12th grade student. So repetition of the same does NOT merit any credit at this stage of education.*

Common mistakes.

1. *A large number of students wrote the following without any justifications*

$$\left(\frac{dy}{dx} - y\right)\left(\frac{dy}{dx} + y\right) = 0 \Longrightarrow \frac{dy}{dx} - y = 0 \text{ or } \frac{dy}{dx} + y = 0.$$

   This is NOT permitted. The same applies to

$$\frac{dy}{dx} = \pm y \Longrightarrow \frac{dy}{dx} = y \text{ or } \frac{dy}{dx} = -y.$$

2. *This is NOT a linear ODE. So linear combination of two solutions need not be a solution. For instance, $e^x - e^{-x}$ is NOT a solution.*

3. *If y' vanishes at a point, say x_0, from that it cannot be concluded that y is a constant.*

4. *$y' = 0$ does NOT imply $y = 0$.*

8. For any $n \in \mathbb{N}$, we let

$$f_n : \mathbb{R} \longrightarrow \mathbb{R},\ f_n(x) \stackrel{\text{def}}{=} \frac{x}{1+nx^2}.$$

Study the uniform convergence of $\{f_n\}_{n=1}^{\infty}$ and $\{f_n'\}_{n=1}^{\infty}$. [5]

Solution. *For any $n \in \mathbb{N}$ and $x \in \mathbb{R}$, it is easy to see that*

$$|f_n(x)| \leqslant \frac{1}{2\sqrt{n}},$$

as $nx^2 - 2\sqrt{n}|x| + 1 = (\sqrt{n}|x| - 1)^2 \geqslant 0$. Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \frac{1}{4\varepsilon^2}$. Then for all $n \geqslant N$ and $x \in \mathbb{R}$, we see that

$$|f_n(x)| \leqslant \frac{1}{2\sqrt{n}} \leqslant \frac{1}{2\sqrt{N}} < \varepsilon.$$

This shows that $\{f_n\}_{n=1}^{\infty}$ converges to the constant 0 function uniformly on $\mathbb{R}$. If $\{f_n'\}_{n=1}^{\infty}$ is uniformly convergent, then one would have $\lim_{n\to\infty} f_n'(0) = 0$, which is not possible as, for all $n \in \mathbb{N}$, $f_n'(0) = 1$. Alternatively, you can show that $\{f_n'\}_{n=1}^{\infty}$ converges pointwise to the following function on $\mathbb{R}$:

$$\begin{cases} 1, & \text{if } x = 0; \\ 0, & \text{if } x \neq 0. \end{cases}$$

Since the poitwise limits of $\{f_n'\}_{n=1}^{\infty}$ is not continuous but each f_n' is continuous, so $\{f_n'\}_{n=1}^{\infty}$ cannot be uniformly convergent.

Comments.

1. *Recall that, if $I \subseteqq \mathbb{R}$ is an open interval, $c \in I$ is a point of local extremum of $\varphi : I \longrightarrow \mathbb{R}$ and φ is differentiable at c, then $\varphi'(c) = 0$. Thus $\varphi'(\xi) = 0$, then ξ might be a point of **local** maximum/minimum. In general, ξ need not be a point of maximum or minimum of φ. Thus, unless there is some other reason, it is NOT mathematically correct to draw the conclusion from $\varphi'(\xi) = 0$ that $\varphi(\xi)$ is is the maximum/minimum of φ on I.*
2. *In order to show the uniform convergence of a sequence of functions $\{g_n\}_{n=1}^{\infty}$, we need to study the behaviour of $\sup_{x\in\mathbb{R}} |g_n(x) - g(x)|$, where is the pointwise limit of $\{g_n\}_{n=1}^{\infty}$. Now even if each g_n and g are differentiable, $|g_n - g|$ need not be differentiable.*
3. *Further to the above, for a given function φ, this is not true always that $|\max \varphi| \geqslant \max |\varphi|$.*

The above thus shows that if the derivative test is use blindfold, it may turn out to be utterly futile.

Common mistakes.

1. *Many students have used the derivative test in order to show the uniform convergence. This is NOT completely correct as mentioned in the comments above.*
2. *From the above comments, it is also clear that even if one finds maximum and minimum of f_n (or f_n'), by the derivative test or otherwise, that does not provide the supremum of $|f_n|$ (or $|f_n'|$ respectively).*

3. *A.M-G.M inequality is to be applied to nonnegative reals.*
4. *The pointwise limit of* $\{f_n'\}_{n=1}^{\infty}$ *is not constantly* 0.

9. Consider the function $f : \mathbb{R} \longrightarrow \mathbb{R}$ defined by

$$f(x) = \sum_{n=1}^{\infty} \frac{|x|}{x^2 + n^2}.$$

(a) Show that f is continuous on $\mathbb{R}$.

Solution. *Let* $R > 0$. *Observe that, for all* $n \in \mathbb{N}$ *and* $x \in [-R, R]$ *one has*

$$\frac{|x|}{x^2 + n^2} \leqslant \frac{R}{n^2 + x^2} \leqslant \frac{R}{n^2}.$$

It follows from Weierstrass M-test that the series $\sum_{n=1}^{\infty} \frac{|x|}{x^2+n^2}$ *converges uniformly on* $[-R, R]$, *since* $\sum_{n=1}^{\infty} \frac{1}{n^2}$ *is convergent. Since each summand is a continuous function, clearly so is* f *on* $[-R, R]$. *As* $R > 0$ *is arbitrary, we conclude that* f *is continuous at every point of* $\mathbb{R}$.

Common mistakes.

1. *This is NOT always true that* $\frac{|x|}{x^2+n^2} \leqslant \frac{1}{n^2}$. *For example, you can take* $x = 2$ *and* $n = 4$.
2. *As* $\sup_{x\in\mathbb{R}} \frac{|x|}{x^2+n^2} = \frac{1}{n}$, *for all* $n \in \mathbb{N}$, *Weierstrass M-test does NOT apply to this case. Also Weierstrass M-test does not apply need not mean that the series cannot converge uniformly.*
3. *Since* $\mathbb{R}$ *is NOT compact, Dini's theorem does not apply here.*

(b) Find all points $x \in \mathbb{R}$ such that f is differentiable at x.

Solution. *Let* $x > 0$. *Observe that, for all* $n \in \mathbb{N}$, *one has*

$$\left|\frac{d}{dx}\left(\frac{x}{x^2 + n^2}\right)\right| = \left|\frac{n^2 - x^2}{(n^2 + x^2)^2}\right| \leqslant \frac{n^2 + x^2}{(n^2 + x^2)^2} \leqslant \frac{1}{n^2 + x^2} \leqslant \frac{1}{n^2}.$$

It follows from Weierstrass M-test that the series $\sum_{n=1}^{\infty} \frac{d}{dx}\left(\frac{x}{x^2+n^2}\right)$ *converges uniformly on* $(0, \infty)$, *since* $\sum_{n=1}^{\infty} \frac{1}{n^2}$ *is convergent. Hence* f *is differentiable on* $(0, \infty)$ *since* $\sum_{n=1}^{\infty} \frac{|x|}{x^2+n^2}$ *converges everywhere. A similar argument yields the differentiability of* f *on* $(-\infty, 0)$. *Finally, we claim that,* f *is not differentiable at* 0. *Observe that, for all* $x \neq 0$,

$$\frac{f(x) - f(0)}{x} = \frac{|x|}{x} \sum_{n=1}^{\infty} \frac{1}{x^2 + n^2}.$$

From this it is clear that $\lim_{x\to 0} \frac{f(x)-f(0)}{x}$ *does not exist as* $\lim_{x\to\infty} \sum_{n=1}^{\infty} \frac{1}{x^2+n^2} = \sum_{n=1}^{\infty} \frac{1}{n^2} > 0$, *due to the uniform convergence of* $\sum_{n=1}^{\infty} \frac{1}{x^2+n^2}$.

Comments. *Uniform convegrence is required here.*

Common mistakes.

1. *Interchange of limit and sum, or derivative and sum CANNOT be done without showing uniform convergence beforehand.*
2. *Some students tried to use Abel's test to establish the uniform convergence of the series* $\sum_{n=1}^{\infty} \frac{|x|}{x^2+n^2}$ *with* $g_n(x) \stackrel{\text{def}}{=} |x|$. *But this is NOT uniformly bounded. So Abel's test cannot be applied with this choice.*

[3+3=6]

10. (a) Let X be a metric space. Assume that X is compact. Show that, any continuous map $f : X \longrightarrow Y$ to any metric space Y is uniformly continuous.

Solution. *Let* $\varepsilon > 0$. *Since* f *is continuous, for all* $x \in X$, *there exists* $\delta_x > 0$ *such that,* $\forall y \in B(x;\delta_x)$, $d(f(y), f(x)) < \frac{\varepsilon}{2}$. *We consider the open cover* $\left\{B\left(x;\frac{\delta_x}{2}\right) : x \in X\right\}$ *of* X. *Since* X *is compact, there exists* $n \in \mathbb{N}$ *such that* $x_1, \ldots, x_n \in X$, $X = \bigcup_{i=1}^{n} B\left(x_i;\frac{\delta_{x_i}}{2}\right)$. *Let* $\delta \stackrel{\text{def}}{=} \min\left\{\frac{\delta_{x_i}}{2}, \ldots, \frac{\delta_{x_n}}{2}\right\}$. *Now consider* $x, y \in X$ *such that* $d(x,y) < \delta$. *Choose* x_i *such that* $x \in B\left(x_i;\frac{\delta_{x_i}}{2}\right)$. *Then*

$$d(y, x_i) \leqslant d(y,x) + d(x, x_i) < \delta + \frac{\delta_{x_i}}{2} \leqslant \frac{\delta_{x_i}}{2} + \frac{\delta_{x_i}}{2} = \delta_{x_i}.$$

It follows that, $d(f(x), f(y)) \leqslant d(f(x), f(x_i)) + d(f(x_i), f(y)) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$.

Comments. *This was proved in the class.*

Common mistakes.

1. *Some students considered an arbitrary cover to begin with, which is not dependent on the function. There is NO reason that such a cover would lead to the correct destination.*
2. *A few started with an arbitrary* $\delta > 0$ *having no dependence with the function. Again, there is no reason that the chosen* $\delta > 0$ *would work.*
3. δ_{x_i}*'s obtained from the open cover* $\{B(x;\delta_x) : x \in X\}$ *and the ones contained from* $\left\{B\left(x;\frac{\delta_x}{2}\right) : x \in X\right\}$ *need not be same.*
4. *For any arbitrary sequence* $\{x_n\}_{n=1}^{\infty}$ *in* X, $B\left(x_n;\frac{\delta_{x_n}}{2}\right)$ *need not be a cover of* X.
5. *One student argues like the following: let* $\varepsilon > 0$. *Then* $\{f^{-1}(B(f(x);\varepsilon) : x \in X\}$ *is an open cover of* X. *Let* $\delta > 0$ *be Lebesgue number corresponding to that cover. Now if* $x, y \in X$ *such that* $d(x,y) < \frac{\delta}{3}$, *then* $B\left(x,\frac{\delta}{3}\right) \subseteq f^{-1}(B(f(x);\varepsilon)$. *This need NOT be true, i.e., need not be the same* x.

(b) Does the converse of (a) hold? Justify your answer.

Solution. *The converse is FALSE.*

Example 1. *Let* $X \stackrel{\text{def}}{=} \mathbb{N}$ *be equipped with the discrete metric. Consider any metric space* Y *and any continuous map* $f : \mathbb{N} \longrightarrow Y$. *Let* $\varepsilon > 0$. *Then, for any* $x, y \in \mathbb{N}$, *with* $|x-y| < 1$, *one clearly has has* $|f(x) - f(y)| = 0 < \varepsilon$.

Example 2. *Let* $X \stackrel{\text{def}}{=} [0,1] \cup \{2\} \cup \{3\} \cup \{4\} \cup \ldots$ *be equipped with the metric induced by the usual metric on* $\mathbb{R}$. *Clearly* X *is not compact as it is not bounded. Consider*

any continuous map $f : X \longrightarrow Y$, where Y is an arbitrary metric space. Suppose that $\varepsilon > 0$. Since f is continuous on $[0,1]$, which is compact, there exists $\delta' > 0$ such that $d(f(x), f(y)) < \varepsilon$, whenever $0 \leqslant x, y \leqslant 1$ and $|x - y| < \delta'$. Let $\delta \stackrel{\text{def}}{=} \min\{\delta', 1\}$. Observe that, if $x, y \in X$ such that $|x - y| < \delta$, then either $x = y$ or $x, y \in [0,1]$. In either case, we must have $|f(x) - f(y)| < \varepsilon$. Hence f is uniformly continuous.

Comments. *This too was mentioned in the class.*

Common mistakes.

1. *$X \stackrel{\text{def}}{=} [0,1] \cup \{1\} \cup \{2\} \cup \{3\}$, equipped with the metric induced by the euclidean metric is compact. This is NOT true that the only subsets of $\mathbb{R}$ that are compact are closed and bounded intervals.*
2. *No matter whatever $n \in \mathbb{N}$ you take, $X = [0,1] \cup \{1\} \cup \{2\} \cup \cdots \cup \{n\}$ is compact.*
3. *Just mentioning the example is NOT enough, justifications must be provided.*

[5+3=8]